Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/16/2016
PI Name:	Snyder, Michael Ph.D.		
Project Title:	HERO Twin Astronaut Study Consortium (TASC) Project: Longitudinal integrated multi-omics analysis of the biomolecular effects of space travel		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305-2200	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/16/2014	**End Date:**	04/15/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Winkelmann, Juliane (Stanford University)		
Grant/Contract No.:	NNX14AH52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Key Objectives: The rigors of space travel present conditions for which our species have not adapted, and thus it would be a significant public health benefit to characterize physiologic and cellular responses to the unique environmental conditions associated with space travel including solar radiation, prolonged weightlessness, G-transitions, and circadian disturbances. Our goal is to perform a longitudinal integrative analysis of all omics data generated during the Human Exploration Research Opportunities (HERO) mission to provide an unprecedented portrait of the molecular and physiological changes that accompany space travel. Methods/Techniques: The Snyder laboratory at Stanford has recently pioneered longitudinal integrative personalized omics profiling (iPOP), which combines genome-wide sequencing with the regular sampling of a patient's transcriptome, proteome, metabolome, and cytokine profiles in order to detect multi-analyte signatures that correlate with disease onset and progression. Our approach will be to leverage our unique expertise in integrated omics studies to combine the data generated at each of the HERO sites in an integrated analysis framework to detect longitudinal multi-analyte signatures that originate specifically during or after the space mission. We hypothesize that through integrative omics analysis we will discover biological signatures associated with G-transitions, prolonged weightlessness, the DNA damage response to solar radiation, and other environmental changes that occur during the space mission. Significance: The proposed work is highly innovative and will provide the first large-scale omics data ever collected on a human during space travel and will present an unprecedented longitudinal analysis of monozygous twins placed in vastly different environments. This comprehensive biological analysis will allow us to better understand the health risks associated with space travel and develop strategies to mitigate these risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Results: Samples on both the inflight and ground subject are currently being collected and analyzed. Discussion: We are currently running biomolecular analyses on the samples acquired so far. Work completed includes genome sequencing of both participants via multiple complementary methods as well as the analysis of circulating biomolecules in the blood. Computational work has included genome and other biomolecular analysis, the development of computational pipelines for integrative data analysis of all Twins Study data as well as the implementation of secure encrypted data transfer and storage solutions for the sharing of datasets across sites. About a terabyte of twins study data have been acquired so far and are securely stored.
Bibliography Type:	Description: (Last Updated: 08/12/2021)
Abstracts for Journals and Proceedings	Piening BD, Salins D, Snyder MP. "Putting it all together: integrative multi -omic analysis in the Twins Study." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Salins D, Piening BD, Snyder MP. "Precision data out of this world management: managing all the omic data generated from the Twins Study." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Rana B, Stenger M, Lee S, Macias B, Siamwala J, Piening B, Hook V, Ebert D, Patel H, Smith S, Snyder M, Hargens A. "Proteomic Assessment of Fluid Shifts and Association with Visual Impairment and Intracranial Pressure in Twin Astronauts." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016